

Literature Seminar (S)

The Big Data Ecosystem:
Foundations, Technologies & Perspectives

Data Engineering I

Course Code: 1TD169

Spring Term 2026 / VT 26

Seminar Information

Points:	1 point (Mandatory)
Team Size:	Same teams as Project
Written Part:	One question per student (+ combined group submission)
Seminar Format:	Cross-team discussion in one seminar session
Deadlines:	See course schedule on Studium

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1 Overview & Purpose

The literature seminar deepens your understanding of the broader big data ecosystem by engaging with primary sources – the original papers that shaped the tools and concepts you encounter in lectures and assignments. The goal is to build a coherent mental model of why things are the way they are: why certain tools exist, what problems they solved, and how they connect to each other.

This understanding directly prepares you for your project, where you will need to make informed architectural decisions – choosing between batch and streaming, storage formats, processing frameworks, and subsequently justifying those decisions in your project report. You will only attend **one** of seminar sessions – the precise schedule will be communicated separately after group formation.

1.1 Learning Objectives

After completing the seminar, you should be able to:

- ✓ Trace the logical evolution of big data processing from MapReduce to modern stream processing and lakehouse architectures
- ✓ Critically compare batch and streaming paradigms and explain when each is appropriate
- ✓ Explain the role of data formats and storage layouts in system performance
- ✓ Situate the tools used in A2 and A3 (Hadoop, Spark) within the broader ecosystem
- ✓ Discuss scalability from both theoretical and practical perspectives
- ✓ Read, cite, and discuss primary technical literature confidently

1.2 Seminar Structure

The seminar consists of two interdependent parts:

1. **Written preparation (individual):** Each team member answers & submits one assigned question (see Section 3). The group also submit a single document covering all questions before the seminar session. Please read each other's texts within the group as seminar preparation.
2. **Seminar session (team):** Teams are grouped together (2 teams per discussion group). You share your written insights with each other, discuss the questions, and learn from the material your teammates prepared. The goal is active dialogue, not presentations.

i Why individual questions for a team seminar?

Each person prepares deeply on one topic, then *teaches* their teammates during the session. This mirrors how real engineering teams work: specialists contribute their expertise to collective decisions. You are responsible for understanding *all five questions* by the end of the session, with your teammates filling in what you did not read.

1.3 Assignment of Questions

Each team member picks one of the five questions in Section 3. For a 5-person team this is exactly one question each. For a 4-person team, you may skip one question and learn about it during the joint discussions. For a 6-person team, one question may be picked by two persons.

2 Reading Material

2.1 How to Use the Papers

The papers listed in Section 7 are your **starting point**, not your complete reading list. You are expected to:

1. Read the papers assigned to your question (see the mapping in Section 2.3)
2. Follow relevant citations in those papers
3. Search for additional literature on your own (Google Scholar, Semantic Scholar, ACM DL, or IEEE Xplore)

i Minimum independent references

At least **two** of the references in your written answer must be to papers *not* provided in the course portal. This is to develop your literature search skills and ensure you go beyond the curated list.

2.2 A Note on Paper Age

Several papers on the reading list are from the mid-2000s to early 2010s. This is intentional: these are the *foundational* works that shaped the field. A practising data engineer who has not read the MapReduce paper or the original Spark paper is missing important intellectual context.

However, the field has progressed significantly. Where relevant to your question, you are encouraged to contrast the original contributions with how they have evolved - or been superseded, as in practice today (based on your additional reading).

2.3 Paper–Question Mapping

The table below provides a suggested starting point for each question. You are not limited to these papers; they can be seen as good anchors.

Question	Suggested starting papers (also presented in References)
Q1 (Batch vs. Streaming)	[Sha14], [Tos+14], [Kul+15], [Car+15], [BHT19]
Q2 (Data Formats)	[Ker+11], [And+13], [CZ14], [ADE+12], [Zen+23]
Q3 (Ecosystem Evolution)	[DG04], [Zah+12], [Zah+10], [Shv+13], [Car+15], [Tos+14], [Kul+15]
Q4 (NoSQL)	[CZ14], [Wu+14], [ADE+12], [Tay10], [McK+10]
Q5 (Scalability & Autoscaling)	[ATE12], [JST21], [LBH13], [ADE+12]

3 Questions

i Written answer guidelines

- **Length:** Approximately half an A4 page per question, font size 12 pt, single spacing.
- **Tone:** Analytical and critical.
- **References:** Cite at least 3–4 sources per question, at least 2 of which must not appear in the course portal reading list.
- **Format:** Consistent citation style throughout (IEEE or ACM recommended). Submit as PDF.
- **Submission:** Upload individually on Studium before the deadline.

? Q1: Batch vs. Streaming Processing

Discuss and contrast the characteristics of batch and streaming data processing, both from a problem perspective and a technology perspective.

Consider: What kinds of applications or data sources *require* streaming? What makes batch processing still relevant and preferable in many contexts? How have frameworks evolved to blur the line between the two paradigms? Discuss concrete examples of tools and their design trade-offs.

Suggested anchors: [Sha14], [Tos+14], [Car+15], [Kul+15]

? Q2: Data Formats and Storage

Discuss the role that data types and formats play in shaping technological solutions for big data processing. Are there specialised tools that achieve high performance or high usability for specific formats?

Consider: Why does the choice of format (row vs. columnar, text vs. binary, schema-on-read vs. schema-on-write) matter for processing performance? How have formats like Parquet and Avro changed the landscape? What does the concept of a *data lakehouse* (e.g., Delta Lake) contribute?

Suggested anchors: [Ker+11], [And+13], [ADE+12] [Zen+23]

? Q3: The Big Data Ecosystem – Is There a Logic?

From a technological point of view, relate the following tools to each other, both historically and in terms of the problems they address. Is there a coherent logical progression in what can appear to be a fragmented ecosystem?

Tools to cover (include others you find interesting):

Google GFS, Apache Hadoop & HDFS, HBase, Google BigTable, Apache Cassandra, Twitter Storm, Twitter Heron, Apache Spark, Apache Kafka, Apache Flink

Consider: What was the original problem each tool aimed to solve? Which tools were reactions to limitations of predecessors? Are there “families” of tools that share design philosophies? Where is the ecosystem heading now (cloud-native, serverless, lakehouse)?

Suggested anchors (choose 3-4 from): [DG04], [Shv+13], [Zah+12], [Zah+10], [Tos+14], [Kul+15], [Car+15]

? Q4: The Rise of NoSQL

Why have NoSQL databases such as MongoDB and Apache Cassandra gained such popularity alongside, and sometimes at the expense of traditional relational databases?

Consider: What are the fundamental trade-offs in the CAP theorem and how do different NoSQL systems navigate them? What workloads or data models are ill-served by relational systems? Are there cases where the NoSQL trend has been overcorrected, and relational databases or NewSQL systems are now returning?

Suggested anchors: [CZ14], [Wu+14], [Tay10], [McK+10]

? Q5: Scalability – Theory and Practice

Scalability is a central concept in distributed systems and a core deliverable in your project. Discuss scalability from both a theoretical basis and from a technological perspective. What is the difference between reactive and proactive autoscaling, and what are the current approaches to achieve it?

Consider: What do Amdahl's Law and Gustafson's Law tell us about the limits of parallelism? What is the difference between strong and weak scaling? What infrastructure mechanisms (YARN, Kubernetes, cloud auto-scaling groups) enable dynamic resource allocation? What are the emerging challenges around autoscaling for ML workloads or edge computing?

Suggested anchors: [ATE12], [JST21], [LBH13], [ADE+12]

? Quick Tip: Connect to Your Project

Every question connects to decisions you will make in your project. As you read, actively ask yourself: “Which of these design choices will I face when building my pipeline?” Note these insights as they will strengthen your project report's motivation and discussion sections.

4 Seminar Session

4.1 Format and Expectations

The seminar session brings together several teams in a discussion group facilitated by an instructor. It is **not** a presentation session; you are not expected to prepare slides. Instead, come prepared to *explain* your question and *engage* with the questions your teammates prepared.

Typical flow:

1. Each person gives a 3–4 minute summary of their question and key findings
2. Open discussion on intersections between questions (e.g., how does Q3 ecosystem evolution relate to Q1 batch vs. streaming?)
3. Instructor-led discussion and summary

4.2 Attendance Policy

Important: Attendance is mandatory

The seminar session is mandatory - you will mark attendance via a QR code. If you cannot attend due to illness or an emergency:

1. Contact an instructor *before* the session
2. Submit a written supplementary assignment (available on Studium in due course)
3. You are still required to submit the written answer to your question and participate in any group report writing

The supplementary assignment substitutes attendance only; all written deliverables remain required.

5 Deliverables and Grading

5.1 What to Submit

Deliverable	Who Submits	Where
Written answer (one question - before seminar) ¹	Individual submission	Studium
Combined Document (all questions - before seminar)	Group submission	Studium
Individual contribution statement (submit after seminar)	Individual submission	Studium

Important: Note the separate individual and group submission

Each student submits their own written answer and their own contribution statement. This is intentional: it ensures individual accountability for the reading and writing work. You also submit a combined document with all answers - this is for the group's benefit, and helps everyone prepare for the seminar by going over all the questions.

5.2 Grading

The seminar is worth **1 point (mandatory)**. To receive this point:

Success Criteria: Seminar Pass Criteria

- ☐ Written answer submitted on time (PDF, correct question, half an A4 page)
- ☐ At least 3–4 references cited; at least 2 are independent (not on the portal reading list)
- ☐ Consistent citation format used throughout
- ☐ Active attendance at seminar session (or supplementary assignment submitted)
- ☐ Individual contribution statement submitted

Seminar is pass/fail

The seminar is assessed as pass/fail only (1 point or 0). There are no advanced points for this component. Focus on quality reading and genuine discussion engagement rather than length or polish.

¹The individual written answer is the primary graded component. All other deliverables (group document, contribution statement) must also be completed to pass.

6 Individual Contribution Statement

Each student submits an individual contribution statement as a separate PDF via Studium **after the seminar**. The purpose is to record what you personally contributed to the seminar preparation, and to acknowledge your learnings from your teammates' contributions.

6.1 Content Requirements

Your statement (maximum one page, 12 pt) must include:

1. **Your full name**
2. **Group number and members**
3. **Seminar participation:** Brief description of your role in the session (what you explained, what you learned from others)
4. **Team acknowledgements:** One take-away point per question, that you learned from your team.
5. **Signature and date**

6.2 Template

Individual Contribution Statement Template

INDIVIDUAL CONTRIBUTION STATEMENT
Literature Seminar (S) – Data Engineering I (1TD169)

Full Name: _____

Group Number: _____

Team Name: _____ **Date:** _____

Team Members: _____

Seminar Participation:
What did you explain to the group? What did you learn from the discussion?

Teammates' Contributions:

- Q__ — *what you learned*
- Q__ — *what you learned*
- Q__ — *what you learned*
- Q__ — *what you learned*

Signature: _____ **Date:** _____

7 Reading List

The following papers and technical reports form the core reading list for this seminar. Papers are grouped thematically. You are strongly encouraged to go beyond this list.

Batch & Streaming Processing

- [DG04] Jeffrey Dean and Sanjay Ghemawat. “MapReduce: Simplified Data Processing on Large Clusters”. In: *Proceedings of the 6th Symposium on Operating Systems Design and Implementation (OSDI '04)*. San Francisco, CA, USA: USENIX Association, 2004, pp. 137–150. URL: <https://research.google/pubs/mapreduce-simplified-data-processing-on-large-clusters/>
- [Zah+12] Matei Zaharia et al. “Resilient Distributed Datasets: A Fault-Tolerant Abstraction for In-Memory Cluster Computing”. In: *Proceedings of the 9th USENIX Conference on Networked Systems Design and Implementation (NSDI '12)*. San Jose, CA, USA: USENIX Association, 2012, pp. 2–2. URL: <https://www.usenix.org/conference/nsdi12/technical-sessions/presentation/zaharia>
- [Zah+10] Matei Zaharia et al. “Spark: Cluster Computing with Working Sets”. In: *Proceedings of the 2nd USENIX Workshop on Hot Topics in Cloud Computing (HotCloud '10)*. Boston, MA, USA: USENIX Association, 2010. URL: https://www.usenix.org/legacy/event/hotcloud10/tech/full_papers/Zaharia.pdf
- [Sha14] Saeed Shahrivari. “Beyond batch processing: towards real-time and streaming big data”. In: *Computers* 3.4 (2014), pp. 117–129. URL: <https://www.mdpi.com/2073-431X/3/4/117>
- [Tos+14] Ankit Toshniwal et al. “Storm@Twitter”. In: *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*. Snowbird, UT, USA: ACM, 2014, pp. 147–156. DOI: [10.1145/2588555.2595641](https://doi.org/10.1145/2588555.2595641)
- [Kul+15] Sanjeeth Kulkarni et al. “Twitter Heron: Stream Processing at Scale”. In: *Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data*. Melbourne, Victoria, Australia: ACM, 2015, pp. 239–250. DOI: [10.1145/2723372.2742788](https://doi.org/10.1145/2723372.2742788)
- [Car+15] Paris Carbone et al. “Apache Flink: Stream and Batch Processing in a Single Engine”. In: *IEEE Data Engineering Bulletin* 38.4 (2015), pp. 28–38. URL: <http://sites.computer.org/debull/A15dec/p28.pdf>
- [BHT19] Ben Blamey, Andreas Hellander, and Salman Toor. “Apache Spark Streaming, Kafka and HarmonicIO: a performance benchmark and architecture comparison for enterprise and scientific computing”. In: *International Symposium on Benchmarking, Measuring and Optimization*. Springer, 2019, pp. 335–347. URL: <https://arxiv.org/pdf/1807.07724>

Storage, Data Formats & Querying

- [Zen+23] Xinyu Zeng et al. “An empirical evaluation of columnar storage formats”. In: *Proceedings of the VLDB Endowment* 17.2 (2023), pp. 148–161. URL: <https://www.vldb.org/pvldb/vol17/p148-zeng.pdf>
- [Shv+13] Konstantin Shvachko et al. *HDFS Architecture Guide*. Tech. rep. Official Apache HDFS architecture documentation. Apache Software Foundation, 2013. URL: <https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HdfsDesign.html>
- [Ker+11] Martin L Kersten et al. “The Researcher’s Guide to the Data Deluge: Querying a Scientific Database in Just a Few Seconds.” In: *Proc. VLDB Endow.* 4.12 (2011), pp. 1474–1477. URL: <https://www.vldb.org/pvldb/vol4/p1474-kersten.pdf>

- [**And+13**] Andrej Andrejev et al. “Scientific analysis by queries in extended SPARQL over a scalable e-Science data store”. In: *2013 IEEE 9th International Conference on e-Science*. IEEE. 2013, pp. 98–106. URL: <https://doi.org/10.1109/eScience.2013.19>

Applications & Domain Use Cases

- [**Nie+12**] Matti Niemenmaa et al. “Hadoop-BAM: Directly Manipulating Next Generation Sequencing Data in the Cloud”. In: *Bioinformatics* 28.6 (2012), pp. 876–877. DOI: [10.1093/bioinformatics/bts054](https://doi.org/10.1093/bioinformatics/bts054)
- [**Tay10**] Ronald C. Taylor. “An Overview of the Hadoop/MapReduce/HBase Framework and Its Current Applications in Bioinformatics”. In: *BMC Bioinformatics* 11.Suppl 12 (2010), S1. DOI: [10.1186/1471-2105-11-S12-S1](https://doi.org/10.1186/1471-2105-11-S12-S1)
- [**McK+10**] Aaron McKenna et al. “The Genome Analysis Toolkit: A MapReduce Framework for Analyzing Next-Generation DNA Sequencing Data”. In: *Genome Research* 20.9 (2010), pp. 1297–1303. DOI: [10.1101/gr.107524.110](https://doi.org/10.1101/gr.107524.110)
- [**Wu+14**] Xindong Wu et al. “Data Mining with Big Data”. In: *IEEE Transactions on Knowledge and Data Engineering* 26.1 (2014), pp. 97–107. DOI: [10.1109/TKDE.2013.109](https://doi.org/10.1109/TKDE.2013.109)

Scalability, Elasticity & Autoscaling

- [**ATE12**] Ahmed Ali-Eldin, Johan Tordsson, and Erik Elmroth. “An adaptive hybrid elasticity controller for cloud infrastructures”. In: *2012 IEEE Network Operations and Management Symposium*. IEEE. 2012, pp. 204–212. URL: <https://doi.org/10.1109/NOMS.2012.6211900>
- [**LBH13**] Monte Lunacek, Jazcek Braden, and Thomas Hauser. “The scaling of many-task computing approaches in python on cluster supercomputers”. In: *2013 IEEE international conference on Cluster computing (cluster)*. IEEE. 2013, pp. 1–8. URL: <https://doi.org/10.1109/CLUSTER.2013.6702678>
- [**JST21**] Li Ju, Prashant Singh, and Salman Toor. “Proactive autoscaling for edge computing systems with kubernetes”. In: *Proceedings of the 14th IEEE/ACM International Conference on Utility and Cloud Computing Companion*. 2021, pp. 1–8. URL: <https://dl.acm.org/doi/abs/10.1145/3492323.3495588>

Surveys & Overviews

- [**CZ14**] C. L. Philip Chen and Chun-Yang Zhang. “Data-Intensive Applications, Challenges, Techniques and Technologies: A Survey on Big Data”. In: *Information Sciences* 275 (2014), pp. 314–347. DOI: [10.1016/j.ins.2014.01.015](https://doi.org/10.1016/j.ins.2014.01.015)
- [**ADE+12**] Divyakant Agrawal, Sudipto Das, Amr El Abbadi, et al. *Challenges and Opportunities with Big Data*. Tech. rep. White paper. Computing Community Consortium (CCC), 2012. URL: <https://cra.org/ccc/wp-content/uploads/sites/2/2015/05/bigdatawhitepaper.pdf>